Hopper Overview

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Agenda

- Who was Hopper?
- Why do we need Hopper?
- Do I need to use Hopper?
- Hopper resources
- How is Hopper like other YCRC clusters?
- How does Hopper differ from other YCRC clusters?
- Resources

Who was Grace Murray Hopper?

- Ph.D. in Mathematics from Yale in 1934
- Pioneering computer scientist
- Trailblazer in creating first compiled computer languages

Why do we need Hopper?

Many datasets now require a higher level of security than Bouchet provides.

- Data Use Agreements (DUAs)
- NIH controlled access datasets (e.g. dbGaP)

Hopper complies with:

- NIST 800-171 standard
- HIPAA standard
- Can meet other specific DUAs

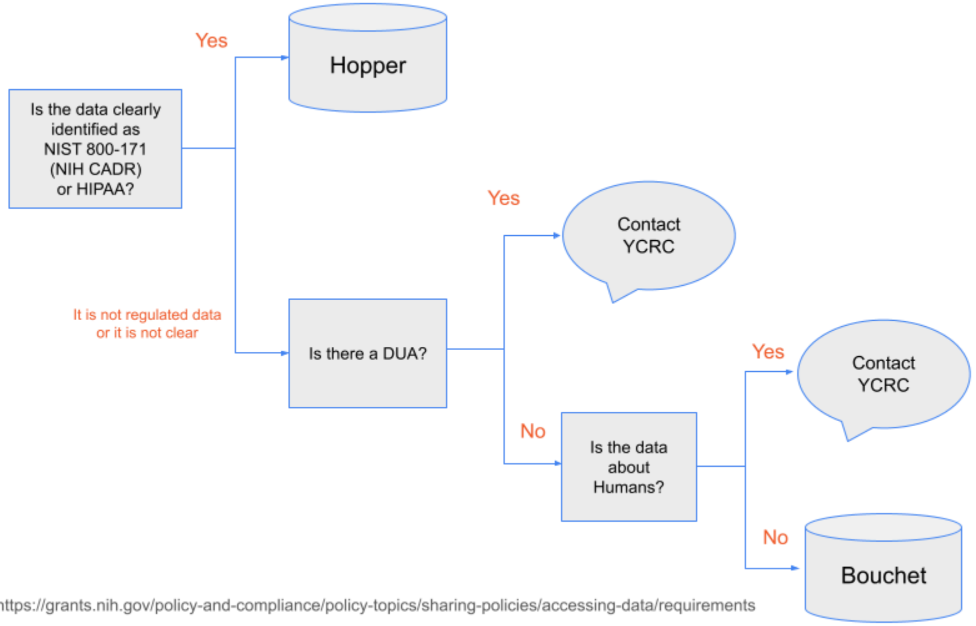
Hopper's Hardware

- ~6000 cpus
- GPUs
  - 40 a5000
  - 40 l40s
  - 40 a40
  - 40 H200
  - 60 H100
  - *48 B200 coming*

VAST 2.4 PiB

https://docs.ycrc.yale.edu/clusters/hopper/#partitions-and-hardware

Do *I* need to use Hopper?



https://grants.nih.gov/policy-and-compliance/policy-topics/sharing-policies/accessing-data/requirements

How is Hopper like other YCRC clusters?

In most ways, Hopper will be familiar to you:

- Slurm jobs
- Partitions: devel, day, week, gpu ...
- Open On Demand (but inside VDI)
- Software modules
- Home, work, scratch directories

How does Hopper differ from other YCRC clusters?

- Getting access
- User Interface (VDI)
- Security measures and rules
- No internet access
- Data transfer
- Software installation
- Usage charges
- Projects, not Groups
- Storage allocations

We'll touch on each in turn

Getting access

Access to the system is a multi-step process:

1. PI submits a project https://research.computing.yale.edu/secure-project-request
2. Health Science IT (HSIT) consults with PI and approves request
3. PI completes required training http://research.computing.yale.edu/regulated-research-training
4. YCRC creates new project, PI account, and associated linux group: *netid_projname*
5. Other lab members can join https://research.computing.yale.edu/hopper-account-request

Of note: a PI can and should submit separate project requests for each individual project (e.g. IRB)

User Interface

- Hopper's only user interface is a Virtual Desktop Infrastructure (VDI)
- You *must* be on the Yale VPN even when on campus
- Two ways to access VDI
  - Via browser: https://hopper.ycrc.yale.edu
  - Via dedicated ThinLinc client. https://www.cendio.com/thinlinc/download
- You will receive a (silent) DUO prompt each time
- The VDI is the equivalent of a login node
- VDI provides terminal, firefox, Open On Demand
- Session is persistent for extdnded period
- Each user has one unique VDI session

Security Measures and Rules

- The VDI prevents copy/pasting with the host computer, prevents file transfers and enforces idle session timeouts.
- Cut/Copy/Paste inside Hopper is allowed (right click + copy/paste)
- External screenshots, screen recording and screen sharing (e.g. via Zoom) are strictly prohibited.
- Internal screenshots can be made with ThinLinc client.
- If you know you will be away from your computer for more than 10 minutes, you must disconnect from the VDI by closing the browser tab or exiting the client.
- You must access Hopper from a private location, such as your home or office. Access from public locations such as coffee shops, transportation hubs or libraries is not allowed.
- Do not put sensitive data (e.g. patient information, personal identifiers) in directory names or job names, which might expose this information.
- Maintenance (usually less than 1 day) is done quarterly.

No Internet Access

- From Hopper, you are not able to directly access anything outside of Hopper.
- All file transfers in and out are mediated by YCRC staff
- With some exceptions, all software installs mediated by YCRC staff

Data Transfer

- Moving data into and out of Hopper requires submitting a request
- We distinquish between Low Risk and Sensitive Data
- Low Risk Transfer In or Out: https://research.computing.yale.edu/hopper-low-risk-transfer : scripts, deidentified data, data files with no PHI, PII, etc.
- Sensitive Data Transfer In: https://research.computing.yale.edu/hopper-sensitive-transfer : everything else
- Sensitive Data Transfer Out: contact YCRC

Low Risk Data Transfer steps

1. Use globus to upload your data to your folder in the "Yale CRC Hopper Low Risk" collection.
2. Submit https://research.computing.yale.edu/hopper-low-risk-transfer
3. YCRC will move your data from the collection to your desired location on Hopper
4. Download is the same in reverse

Sensitive Data Transfer In

1. Submit https://research.computing.yale.edu/hopper-sensitive-transfer first.
2. YCRC will approve, make all necessary networking changes, and create a temporary globus receiving collection
3. You will receive instructions on how to transfer your data.

Transfers requiring method other than globus are possible with YCRC assistance.

Sensitive Data Transfer Out

Transfer of sensitive data out is currently handled on a case-by-case basis. Please contact YCRC.

Software Installation

- Since you have no internet access, most software installation must be done by YCRC.
- The applications will generally be made available as modules.

There are two exceptions allowing self-service:

- python packages from the default anaconda repo. Note: NOT conda-forge

```
module load miniconda
conda create/install ...
```

- many approved R packages from CRAN. Use `install.packages()`

For all other software, first search the modules, then contact YCRC.

What about your own code?

You may transfer in scripts. You may transfer in and compile code written by you or others in your lab. This is considered a low risk transfer.

You may NOT transfer in and compile code written elsewhere. Ask us to download and build it for you.

Usage Charges

Unlike YCRC's other clusters, currently *all* use of Hopper incurs compute charges.

Current Rates:

type	cost/hour
CPUs	$0.004
mid-range GPU	$0.49
H200/B200	$0.99-1.49

Storage beyond free quota: $5.15/TiB/month

See https://docs.ycrc.yale.edu/clusters/hopper/#rate-structure

Projects, not Groups

Accounts on Hopper are different from other YCRC clusters

- Each logical *Project* is set up independently, with its own linux group and membership
- Projects are intended to be impermanent. Once the research project is complete, the project is terminated
- Projects must be reauthorized annually.
- PIs can and should have multiple Hopper projects if working on unrelated research projects
- Each project has its own work and scratch directories
- However, each user has a single home directory

Storage Allocations

space	size	snapshots
home	125 GiB	7 days
work	1 TiB	7 days
scratch	10 TiB	NO
PI	charged	NO

Aside from the snapshots, *no data is backed up!*

Getting help

- Hopper doc https://docs.ycrc.yale.edu/clusters/hopper
- Reach out to YCRC: email ycrc@yale.edu
- Hopper-specific Office Hours https://yale.zoom.us/my/ycrcsupport: Thurs 2-3pm
- *Please do not send us screen shots!*
- We cannot view your screen over zoom. Instead, we can join your VDI session

Workshop Feedback

Please help us improve this workshop by sharing feedback via a 2-minute anonymous survey. Thank you.

https://yalesurvey.ca1.qualtrics.com/jfe/form/SV_dgL2LLO0KpHWYqq

Scan the QR Code:

Questions?

Thank you!

Any remaining time can be used for additional questions and office hours.